

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
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| 1(a)    | light / optical (microscope) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |
| 1(b)    | 460 / 464 ( $\mu\text{m}$ ) ; <b>A</b> 450–480                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>1</b> |
| 1(c)    | <p>any <b>one</b> from:</p> <p>increase the surface area ;</p> <p>reach a larger volume of, soil / soil particles ;</p> <p>AVP ;    e.g. (root hairs) increase the number of aquaporins in cell surface membrane</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>1</b> |
| 1(d)    | <p>max <b>two</b> if answer only refers to either <math>\text{Na}^+</math> or <math>\text{K}^+</math> ions</p> <p>any <b>three</b> from:</p> <p>1 facilitated diffusion because, higher concentration <math>\text{Na}^+</math> in soil solution / lower concentration <math>\text{Na}^+</math> in root hair cells ;</p> <p>2 active, transport / uptake, because, higher concentration <math>\text{K}^+</math> in root hair cells / lower concentration <math>\text{K}^+</math> in soil solution ;</p> <p>3 ions / AW, cannot, diffuse / pass through the, cell (surface) membrane / phospholipid bilayer ;</p> <p>4 (so) facilitated diffusion uses, channel / carrier proteins</p> <p><b>or</b></p> <p>active transport uses carrier proteins ;</p>                                                                                                                                              | <b>3</b> |
| 1(e)    | <p>(movement of water) from cell <b>A</b> to cell <b>B</b> by osmosis ;</p> <p>from higher water potential to lower water potential / from less negative <math>\psi</math> to more negative <math>\psi</math> / down the water potential gradient ;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>2</b> |
| 1(f)(i) | <p>any <b>four</b> from:</p> <p>1 (carbon dioxide reacts with water resulting in) carbonic acid produced ;</p> <p><b>accept</b> from equation</p> <p>2 carbonic acid dissociates to form, hydrogencarbonate (ions) / <math>\text{HCO}_3^-</math>, and, protons / <math>\text{H}^+</math> ;</p> <p><b>accept</b> from equation</p> <p><b>accept</b> bicarbonate for hydrogencarbonate</p> <p>3 pH of soil (solution) lowered / soil (solution) becomes more acidic ;</p> <p><b>accept</b> correct paired data quotes as alternative to lower pH</p> <p>4 at lower pH clay particles lose <math>\text{Fe}^{2+}</math> / <math>\text{H}^+</math> displaces <math>\text{Fe}^{2+}</math> from clay particles ;</p> <p>5 hydrogencarbonate ions attract (bound) <math>\text{Fe}^{2+}</math> into soil solution ;</p> <p>6 AVP ;</p> <p>e.g. no carbonic anhydrase in soil solution, so reaction slow</p> | <b>4</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
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| 1(f)(ii) | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 (Fe<sup>2+</sup>) must cross the cell surface membrane at endodermis / Fe<sup>2+</sup> moves into, cytoplasm / symplastic route ;</li> <li>2 due to Casparian strip / impermeability of suberin / AW ;</li> <li>3 idea that the cell regulates the, quantity / activity, of transport proteins ;</li> <li>4 metabolised / changed, to insoluble iron compounds (and stored in root) ;</li> <li>5 transported into / stored in, vacuole of endodermal cell ;</li> </ol> | 3     |

| Question | Answer | Marks |
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